

Lecture Notes: Derivatives

Calculus I

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Section 1 The Definition of the Derivative

Definition 1 (Derivative at a Point) The derivative of a function f at a number a , denoted by $f'(a)$, is defined as

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists. If the limit exists, we say f is differentiable at a . If the limit does not exist, we say f is not differentiable at a .

Definition 2 (Derivative Function) The derivative function f' is defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

for all x for which the limit exists. The domain of f' consists of all numbers in the domain of f where f is differentiable.

Remark 1 (Notation) Several notations are commonly used for the derivative:

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{df}{dx}, \quad D_x f(x), \quad Df(x).$$

The notation $\frac{dy}{dx}$ is called Leibniz notation and is particularly useful when dealing with differentials and implicit differentiation.

Example 1 Compute the derivative of $f(x) = x^2$ using the definition.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

Thus, $f'(x) = 2x$.

Example 2 Compute the derivative of $f(x) = \sqrt{x}$ using the definition.

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

Multiply numerator and denominator by the conjugate $\sqrt{x+h} + \sqrt{x}$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

Thus, $f'(x) = \frac{1}{2\sqrt{x}}$ for $x > 0$.

Theorem 1 (Differentiability Implies Continuity) If f is differentiable at a , then f is continuous at a .

Suppose f is differentiable at a . Then

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \cdot (x - a) \right) = \left(\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right) \cdot \left(\lim_{x \rightarrow a} (x - a) \right) = f'(a) \cdot 0 = 0.$$

Therefore, $\lim_{x \rightarrow a} f(x) = f(a)$, so f is continuous at a .

Remark 2 The converse of the theorem is false. The function $f(x) = |x|$ is continuous at 0 but not differentiable at 0. Indeed,

$$\lim_{h \rightarrow 0^+} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1, \quad \text{but} \quad \lim_{h \rightarrow 0^-} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1.$$

Since the left and right limits differ, $f'(0)$ does not exist.

Section 2 Basic Differentiation Rules

Theorem 2 (Constant Rule) If $f(x) = c$ for a constant c , then $f'(x) = 0$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0.$$

Theorem 3 (Power Rule) If $f(x) = x^n$ for a positive integer n , then $f'(x) = nx^{n-1}$.

Using the binomial theorem,

$$(x + h)^n = x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + h^n.$$

Then

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x + h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + h^n}{h} = \lim_{h \rightarrow 0} \left(nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + h^{n-1} \right) = nx^{n-1}.$$

Theorem 4 (Constant Multiple Rule) If c is a constant and f is differentiable, then

$$\frac{d}{dx}[cf(x)] = cf'(x).$$

Theorem 5 (Sum and Difference Rules) If f and g are differentiable, then

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x), \quad \frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x).$$

Theorem 6 (Product Rule) If f and g are differentiable, then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

Let $F(x) = f(x)g(x)$. Then

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

Add and subtract $f(x+h)g(x)$ in the numerator:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left(f(x+h) \frac{g(x+h) - g(x)}{h} + g(x) \frac{f(x+h) - f(x)}{h} \right) = f(x)g'(x) + g(x)f'(x). \end{aligned}$$

Theorem 7 (Quotient Rule) If f and g are differentiable and $g(x) \neq 0$, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}.$$

Let $F(x) = \frac{f(x)}{g(x)}$. Then

$$F'(x) = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)}.$$

Add and subtract $f(x)g(x)$ in the numerator:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\ &= \lim_{h \rightarrow 0} \frac{g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h}}{g(x+h)g(x)} = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}. \end{aligned}$$

Example 3 Differentiate $f(x) = 3x^4 - 2x^2 + 5x - 7$.

$$f'(x) = 3(4x^3) - 2(2x) + 5(1) - 0 = 12x^3 - 4x + 5.$$

Example 4 Differentiate $f(x) = (x^2 + 1)(x^3 - 2x)$. Using the product rule with $u = x^2 + 1$ and $v = x^3 - 2x$:

$$f'(x) = (2x)(x^3 - 2x) + (x^2 + 1)(3x^2 - 2) = 2x^4 - 4x^2 + 3x^4 - 2x^2 + 3x^2 - 2 = 5x^4 - 3x^2 - 2.$$

Example 5 Differentiate $f(x) = \frac{x^2+1}{x-1}$. Using the quotient rule with $u = x^2 + 1$ and $v = x - 1$:

$$f'(x) = \frac{(2x)(x-1) - (x^2+1)(1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}.$$

Section 3 Derivatives of Trigonometric Functions

Theorem 8 (Derivative of Sine and Cosine)

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x.$$

We use the definition and the trigonometric identity $\sin(x+h) = \sin x \cos h + \cos x \sin h$.

$$\begin{aligned} \frac{d}{dx}(\sin x) &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \left(\sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h} \right) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x. \end{aligned}$$

The derivative of cosine is similar, using $\cos(x+h) = \cos x \cos h - \sin x \sin h$.

Theorem 9 (Derivatives of Other Trigonometric Functions)

$$\begin{aligned} \frac{d}{dx}(\tan x) &= \sec^2 x, & \frac{d}{dx}(\cot x) &= -\csc^2 x, \\ \frac{d}{dx}(\sec x) &= \sec x \tan x, & \frac{d}{dx}(\csc x) &= -\csc x \cot x. \end{aligned}$$

For $\tan x = \frac{\sin x}{\cos x}$, use the quotient rule:

$$\frac{d}{dx}(\tan x) = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

The others follow similarly.

Example 6 Differentiate $f(x) = x^2 \sin x$. Using the product rule:

$$f'(x) = 2x \sin x + x^2 \cos x.$$

Example 7 Differentiate $f(x) = \frac{\sin x}{1+\cos x}$. Using the quotient rule:

$$f'(x) = \frac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \frac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \frac{\cos x + 1}{(1+\cos x)^2} = \frac{1}{1+\cos x}.$$

Section 4 Derivatives of Transcendental Functions

Theorem 10 (Derivative of Exponential Functions)

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(a^x) = a^x \ln a \quad (a > 0).$$

For e^x , use the definition:

$$\frac{d}{dx}(e^x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x.$$

For $a^x = e^{x \ln a}$, use the chain rule:

$$\frac{d}{dx}(a^x) = \frac{d}{dx}(e^{x \ln a}) = e^{x \ln a} \cdot \ln a = a^x \ln a.$$

Theorem 11 (Derivative of Logarithmic Functions)

$$\frac{d}{dx}(\ln x) = \frac{1}{x}, \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}.$$

Let $y = \ln x$. Then $e^y = x$. Differentiate implicitly with respect to x :

$$e^y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}.$$

For $\log_a x = \frac{\ln x}{\ln a}$, we get $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$.

Theorem 12 (Derivatives of Inverse Trigonometric Functions)

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}, \quad \frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}.$$

Let $y = \arcsin x$. Then $\sin y = x$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. Differentiate implicitly:

$$\cos y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\cos y}.$$

Since $\cos y \geq 0$ on the range of $\arcsin$, $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Thus $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$. The others are similar.

Example 8 Differentiate $f(x) = e^{2x} \ln x$. Using the product rule:

$$f'(x) = 2e^{2x} \ln x + e^{2x} \cdot \frac{1}{x} = e^{2x} \left(2 \ln x + \frac{1}{x} \right).$$

Example 9 Differentiate $f(x) = \arctan(x^2)$. Using the chain rule:

$$f'(x) = \frac{1}{1+(x^2)^2} \cdot 2x = \frac{2x}{1+x^4}.$$

Section 5 The Chain Rule

Theorem 13 (Chain Rule) *If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x , and*

$$F'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Let $u = g(x)$. We want to find $\frac{dy}{dx}$ where $y = f(u)$. Using the definition,

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \right).$$

As $\Delta x \rightarrow 0$, $\Delta u \rightarrow 0$ because g is continuous (since it is differentiable). Thus,

$$\frac{dy}{dx} = \left(\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \cdot \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right) = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Example 10 Differentiate $f(x) = (3x^2 + 1)^5$. Let $u = 3x^2 + 1$. Then $f(x) = u^5$.

$$f'(x) = 5u^4 \cdot \frac{du}{dx} = 5(3x^2 + 1)^4 \cdot 6x = 30x(3x^2 + 1)^4.$$

Example 11 Differentiate $f(x) = \sin(e^{x^2})$. Let $u = e^{x^2}$. Then $f(x) = \sin u$.

$$f'(x) = \cos u \cdot \frac{du}{dx} = \cos(e^{x^2}) \cdot e^{x^2} \cdot 2x = 2xe^{x^2} \cos(e^{x^2}).$$

Example 12 Differentiate $f(x) = \ln(\sqrt{x} + 1)$.

$$f'(x) = \frac{1}{\sqrt{x} + 1} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}(\sqrt{x} + 1)}.$$

Section 6 Second and Higher Order Derivatives

Definition 3 (Higher Order Derivatives) *The second derivative of f is the derivative of f' , denoted by f'' or $\frac{d^2y}{dx^2}$. In general, the n -th derivative of f is denoted by $f^{(n)}$ or $\frac{d^ny}{dx^n}$.*

Example 13 Find all derivatives of $f(x) = x^5 - 3x^3 + 2x$.

$$\begin{aligned} f'(x) &= 5x^4 - 9x^2 + 2, \\ f''(x) &= 20x^3 - 18x, \\ f'''(x) &= 60x^2 - 18, \\ f^{(4)}(x) &= 120x, \\ f^{(5)}(x) &= 120, \\ f^{(6)}(x) &= 0. \end{aligned}$$

Example 14 Find the second derivative of $f(x) = \sin x$.

$$f'(x) = \cos x, \quad f''(x) = -\sin x.$$

Remark 3 *In physics, if $s(t)$ is the position of an object at time t , then $s'(t)$ is the velocity $v(t)$ and $s''(t)$ is the acceleration $a(t)$.*

Section 7 Implicit Differentiation

Definition 4 (Implicit Differentiation) When a function is defined implicitly by an equation relating x and y , we can find $\frac{dy}{dx}$ by differentiating both sides of the equation with respect to x and then solving for $\frac{dy}{dx}$. This technique is called implicit differentiation.

Example 15 Find $\frac{dy}{dx}$ if $x^2 + y^2 = 25$. Differentiate both sides with respect to x :

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

Example 16 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$. From the previous example, the slope is $\frac{dy}{dx} = -\frac{x}{y} = -\frac{3}{4}$. The tangent line is $y - 4 = -\frac{3}{4}(x - 3)$, or $3x + 4y = 25$.

Example 17 Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$. Differentiate both sides:

$$3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}.$$

Solve for $\frac{dy}{dx}$:

$$3y^2 \frac{dy}{dx} - 6x \frac{dy}{dx} = 6y - 3x^2 \implies \frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

Section 8 Linearization and Differentials

Definition 5 (Linearization) The linearization of f at a is the linear function

$$L(x) = f(a) + f'(a)(x - a).$$

The approximation $f(x) \approx L(x)$ for x near a is called the linear approximation or tangent line approximation.

Definition 6 (Differentials) Let $y = f(x)$ be a differentiable function. The differential dx is an independent variable. The differential dy is defined as

$$dy = f'(x) dx.$$

Example 18 Use linearization to approximate $\sqrt{4.1}$. Let $f(x) = \sqrt{x}$ and $a = 4$. Then $f(4) = 2$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$.

$$L(x) = 2 + \frac{1}{4}(x - 4).$$

For $x = 4.1$:

$$\sqrt{4.1} \approx L(4.1) = 2 + \frac{1}{4}(0.1) = 2 + 0.025 = 2.025.$$

The actual value is approximately 2.02484567, so the approximation is quite accurate.

Example 19 The radius of a sphere is measured to be 5 cm with a possible error of 0.1 cm. Estimate the maximum error in the calculated volume. The volume is $V = \frac{4}{3}\pi r^3$. The differential is

$$dV = 4\pi r^2 dr.$$

With $r = 5$ and $dr = 0.1$:

$$dV = 4\pi(5)^2(0.1) = 4\pi(25)(0.1) = 10\pi \approx 31.4 \text{ cm}^3.$$

Section 9 Derivative of Parametric Functions

Definition 7 (Parametric Curves) A parametric curve is defined by equations

$$x = f(t), \quad y = g(t)$$

where t is a parameter. The derivative $\frac{dy}{dx}$ is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \text{provided } \frac{dx}{dt} \neq 0.$$

By the chain rule, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$. Solving for $\frac{dy}{dx}$ gives the result.

Theorem 14 (Second Derivative of Parametric Curves)

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}.$$

Example 20 Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for the curve $x = t^2$, $y = t^3$.

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3t^2 \implies \frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2} \quad (t \neq 0).$$

For the second derivative:

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{3t}{2} \right) = \frac{3}{2}, \quad \frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}.$$

Example 21 Find the equation of the tangent line to the curve $x = \cos t$, $y = \sin t$ at $t = \frac{\pi}{4}$.

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t \implies \frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t.$$

At $t = \frac{\pi}{4}$, the point is $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$ and the slope is $-\cot \frac{\pi}{4} = -1$. The tangent line is $y - \frac{\sqrt{2}}{2} = -1 \left(x - \frac{\sqrt{2}}{2} \right)$, or $x + y = \sqrt{2}$.

Section 10 Tangents and Normal Lines

Definition 8 (Tangent Line) The tangent line to the curve $y = f(x)$ at the point $(a, f(a))$ is the line with slope $f'(a)$:

$$y - f(a) = f'(a)(x - a).$$

Definition 9 (Normal Line) The normal line to the curve $y = f(x)$ at the point $(a, f(a))$ is the line perpendicular to the tangent line at that point. Its slope is $-\frac{1}{f'(a)}$ (provided $f'(a) \neq 0$):

$$y - f(a) = -\frac{1}{f'(a)}(x - a).$$

Example 22 Find the equations of the tangent and normal lines to $f(x) = x^2$ at $x = 1$. $f(1) = 1$, $f'(x) = 2x$, so $f'(1) = 2$. Tangent line: $y - 1 = 2(x - 1) \implies y = 2x - 1$. Normal line: $y - 1 = -\frac{1}{2}(x - 1) \implies y = -\frac{1}{2}x + \frac{3}{2}$.

Example 23 Find the points on the curve $y = x^3 - 3x^2 + 2x$ where the tangent line is horizontal. The tangent line is horizontal when $y' = 0$.

$$y' = 3x^2 - 6x + 2.$$

Set $y' = 0$: $3x^2 - 6x + 2 = 0$. Using the quadratic formula,

$$x = \frac{6 \pm \sqrt{36 - 24}}{6} = \frac{6 \pm \sqrt{12}}{6} = \frac{6 \pm 2\sqrt{3}}{6} = 1 \pm \frac{\sqrt{3}}{3}.$$

The corresponding y -values can be found by substitution.

Example 24 Find the equation of the normal line to $y = \ln x$ at $x = e$. $f(e) = \ln e = 1$. $f'(x) = \frac{1}{x}$, so $f'(e) = \frac{1}{e}$. The slope of the normal line is $-e$. The normal line is $y - 1 = -e(x - e) \implies y = -ex + e^2 + 1$.